Standard Operating Procedure (SOP)

OpenShift 4.20 Disconnected Mirror Registry Preparation

1. System Prerequisites

Prepare a bastion server running **RHEL 9**. Ensure the server has internet access (or proxy access) for the initial download phase. You must install the foundational container and utility packages first:

```
sudo dnf install -y podman jq openssl tar wget
```

2. Download & Extract Binaries

Download the OpenShift 4.20 specific binaries and the Quay mirror registry package. Extract them and place them in the system path.

```
tar -xzvf openshift-client-linux-4.20.tar.gz -C /usr/local/bin oc kubectl
tar -xzvf openshift-install-linux.tar.gz -C /usr/local/bin openshift-install
tar -xzvf oc-mirror.rhel9.tar.gz -C /usr/local/bin/
chmod +x /usr/local/bin/oc /usr/local/bin/kubectl /usr/local/bin/openshift-install /usr/local/bin/oc-mirror
```

3. Registry Setup & Installation

Create the required mount directories, configure your proxy bypass, and install the Quay registry. This command will auto-generate a secure password for the `init` user.

```
cd /mirror-registry
mkdir -p config storage

export no_proxy=<<FQDN>>,127.0.0.1,localhost
# Note: Ensure <<FQDN>> is added to /etc/hosts resolving to the local IP

./mirror-registry install   --quayHostname <<FQDN>>   --quayRoot /mirror-registry/config   --quayStorage /mirror-r
```

** Keep the generated password handy from the installation output for Step 5.*

4. SSL Generation & OS Trust

Generate a 10-year SSL certificate with a Subject Alternative Name (SAN), assign it to Quay, and crucially, **trust the Root CA at the OS level** so `podman` and `oc-mirror` can authenticate locally.

```
mkdir -p /mirror-registry/newca && cd /mirror-registry/newca
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -nodes -key rootCA.key -sha256 -days 3650 -out rootCA.pem -subj "/CN=MyRootCA"
openssl genrsa -out ssl.key 2048

cat <<EOF > openssl.cnf
[req]
distinguished_name = req_distinguished_name
req_extensions = v3_req
prompt = no
[req_distinguished_name]
CN = <<FQDN>>
[v3_req]
basicConstraints = CA:FALSE
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
subjectAltName = @alt_names
[alt_names]
```

```
DNS.1 = <<FQDN>>
EOF

openssl req -new -key ssl.key -out ssl.csr -reqexts v3_req -config openssl.cnf
openssl x509 -req -in ssl.csr -CA rootCA.pem -CAkey rootCA.key -CAcreateserial   -out ssl.cert -days 3650 -extensi

# Assign to Quay
systemctl stop quay-app.service
cp ssl.cert ssl.key /mirror-registry/config/quay-config/
chown -R 1001:1001 /mirror-registry/config/quay-config/
systemctl start quay-app.service

# Trust the Root CA on the Bastion Host
cp /mirror-registry/newca/rootCA.pem /etc/pki/ca-trust/source/anchors/
update-ca-trust extract
```

5. Pull Secret Configuration & Hygiene

CRITICAL SECURITY REQUIREMENT: The merged pull secret below contains local Quay write credentials (`init` user). It must **ONLY** be used by the bastion for `oc-mirror`. **NEVER** inject this file into the cluster's `install-config.yaml`.

```
# Generate base64 auth token for local Quay using the auto-generated password
echo -n 'init:<<AUTO_GENERATED_PASSWORD>>' | base64 -w0

# Merge Red Hat pull secret (pull-secret.txt) with local Quay auth
cat pull-secret.txt | jq '.auths += {"<<FQDN>>:8443": {"auth": "<<BASE64_TOKEN>>", "email": "systemadmin@xyz.co.in

# Configure authentication file for oc-mirror and podman
mkdir -p $XDG_RUNTIME_DIR/containers
cp /root/pull-secret-mirroring.json $XDG_RUNTIME_DIR/containers/auth.json

# Verify podman login works
podman login -u init -p <<AUTO_GENERATED_PASSWORD>> <<FQDN>>:8443
```

6. Image Mirroring via oc-mirror v2

Use the OpenShift 4.20 declarative mirror workflow (`ImageSetConfiguration`) to synchronize the required cluster images.

1. Create the `imageset-config.yaml` file:

```
apiVersion: mirror.openshift.io/v2alpha1
kind: ImageSetConfiguration
mirror:
  platform:
    channels:
      - name: stable-4.20
        minVersion: 4.20.0
        maxVersion: 4.20.0
    graph: true
```

2. Execute dry-run to validate configuration:

```
oc-mirror -c imageset-config.yaml docker://<<FQDN>>:8443/ocp4 --v2 --workspace file:///mirror-registry/oc-mirro
```

3. Run the complete mirror operation:

```
oc-mirror -c imageset-config.yaml docker://<<FQDN>>:8443/ocp4 --v2 --workspace file:///mirror-registry/oc-mirro
```

Once complete, the `oc-mirror` tool will automatically generate the `ImageDigestMirrorSet` (IDMS) and `ImageTagMirrorSet` (ITMS) manifests inside the `/mirror-registry/oc-mirror-workspace/working-dir/cluster-resources/` directory. You will use these generated files in the OpenShift installer.